

Assignment 1: Python Programming

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1 Variables

1.1 What is a Variable?

A variable is a name used to store data in a program. It acts like a container that holds values such as numbers or text. In Python, a variable is created when a value is assigned to it.

```
[language=Python] name = "Abhijeet" age = 22
```

1.2 Why do We Need Variables?

Variables help us store and reuse data in a program. They make the code easy to read, update, and manage without writing the same value repeatedly.

1.3 Where and How is it Used?

Variables are used in almost every Python program to store information like names, marks, or calculation results. We create a variable using the assignment operator (=) and use its name whenever we need the stored value.

```
[language=Python] name = "Abhijeet" age = 22  
print(name) print(age)
```

2 Basic Data Types

2.1 Integer

What is an Integer?

An integer is a whole number without a decimal point. It can be positive, negative, or zero.

```
[language=Python] age = 22 marks = 95
```

Why do We Need Integers?

Integers are used to store whole numbers such as age, marks, quantity, and count.

Where and How is it Used?

Integers are used in programs for counting and calculations. We store a whole number in a variable and use it whenever needed.

```
[language=Python] age = 22 print(age)
```

2.2 Float

What is a Float?

A float is a number that contains a decimal point.

```
[language=Python] cgpa = 8.45 price = 99.99
```

Why do We Need Floats?

Floats are used to store decimal values such as CGPA, temperature, price, and height.

Where and How is it Used?

Floats are used in programs where accurate decimal values are required.

```
[language=Python] cgpa = 8.45 print(cgpa)
```

2.3 Boolean

What is a Boolean?

A Boolean data type stores only two values: True and False.

```
[language=Python] isStudent = True
```

Why do We Need Boolean?

Boolean values are used to check conditions and make decisions in a program.

Where and How is it Used?

Boolean values are commonly used with if statements and loops.

```
[language=Python] isStudent = True print(isStudent)
```

2.4 String

What is a String?

A string is a sequence of characters enclosed in single or double quotes.

```
[language=Python] name = "Abhijeet"
```

Why do We Need Strings?

Strings are used to store text such as names, addresses, and messages.

Where and How is it Used?

Strings are used whenever a program needs to store or display text.

3 Arithmetic Operators

3.1 What are Arithmetic Operators?

Arithmetic operators are used to perform mathematical calculations in Python. They work with numbers and return the calculated result.

Common Arithmetic Operators

Operator	Meaning
+	Addition
-	Subtraction
	Multiplication
/	Division
//	Floor Division
%	Modulus
*	Exponent (Power)

3.2 Why do We Need Arithmetic Operators?

Arithmetic operators help us perform calculations such as addition, subtraction, multiplication, and division. They make mathematical operations simple and fast in a program.

3.3 How is it Used?

Arithmetic operators are used by writing the operator between two values or variables. Python performs the calculation and returns the result.

4 Type Conversion

4.1 What is Type Conversion?

Type conversion is the process of changing one data type into another. In Python, we can convert data types using built-in functions such as `int()`, `float()`, `str()`, and `bool()`.

Common Type Conversion Functions

Function	Converts To
<code>int()</code>	Integer
<code>float()</code>	Float
<code>str()</code>	String
<code>bool()</code>	Boolean

4.2 Why do We Need Type Conversion?

Type conversion helps us convert data into the required type before performing operations. It allows different data types to work correctly in a program.

4.3 How is it Used?

Type conversion is done by passing a value to a conversion function. Python converts the value to the specified data type.

5 Input and Output

5.1 What is Input and Output?

Input is the data entered by the user into a program. Output is the result displayed by the program after processing the input. In Python, the `input()` function is used to take input, and the `print()` function is used to display output.

5.2 Why do We Need Input and Output?

Input allows users to provide data to the program, while output displays the results. They make programs interactive and user-friendly.

5.3 How is it Used?

The `input()` function takes input from the user and stores it in a variable. The `print()` function displays the value stored in the variable or the result of an operation.

6 Strings

6.1 What is a String?

A string is a sequence of characters enclosed in single quotes (`' '`) or double quotes (`" "`). It is used to store text such as names, messages, and addresses.

6.2 Why do We Need Strings?

Strings are used to store and display text in a program. They help us work with names, messages, passwords, and other text data.

6.3 How is it Used?

Strings are created by writing text inside quotes and storing it in a variable. We can access characters, combine strings, and perform different operations on them.

6.4 String Indexing

String indexing is used to access individual characters of a string. The first character has index 0, and the last character can be accessed using -1.

6.5 String Slicing

String slicing is used to extract a part of a string. It is done using the syntax `string[start:end]`.

6.6 Common String Operations

Some common string operations are concatenation, repetition, changing case, and finding the length of a string.

7 Lists

7.1 What is a List?

A list is a collection of items stored in a single variable. Lists are created using square brackets [] and can store different types of data.

7.2 Why do We Need Lists?

Lists help us store multiple values in a single variable. They make it easy to organize, access, and update data.

7.3 How is it Used?

A list is created using square brackets and its elements can be accessed using their index number.

```
print(fruits[2])
```

7.4 List Indexing

List indexing is used to access individual elements of a list. The first element has index 0 and the last element can be accessed using -1.

7.5 List Slicing

List slicing is used to access a range of elements from a list.

7.6 Updating List Elements

List elements can be changed by assigning a new value to their index.

7.7 Nested Lists

A nested list is a list that contains another list as an element.

7.8 Common List Methods

append()

Adds an element at the end of the list.

insert()

Inserts an element at a specified position.

remove()

Removes the specified element from the list.

pop()

Removes and returns the last element by default.

sort()

Sorts the list in ascending order.

reverse()

Reverses the order of the list.

len()

Returns the number of elements in a list.